

ENGINEERING REFLECTION:

1. Source profiling should happen before ingestion because it reveals data quality issues early. By knowing about duplicates, missing values, or inconsistent formats, we can design validation rules and avoid surprises during pipeline execution.
2. Source event time or `updated_at` refers to when the data was created or changed in the source system. Ingestion time is when our pipeline actually loads the record. They are different because ingestion can lag behind the source, and we rely on `updated_at` to know what is new.
3. `Event_id` alone is insufficient because the same `event_id` can appear multiple times with different `updated_at` values. We need both `event_id` and `updated_at` to decide which version is the freshest record.
4. The watermark must only advance after successful persistence. If we advance it too early and the write fails, we would skip those records forever on the next run, causing data loss.
5. Using `updated_at > watermark` has a limitation when multiple records share the same timestamp. Records equal to the watermark might be missed, so inclusive comparisons and deduplication are safer.
6. Duplicate prevention is directly related to idempotency. If we can rerun the pipeline without creating duplicates, then the pipeline is idempotent. Deduplication ensures each record is represented once even across multiple runs.
7. The raw area should preserve source values exactly as they are. Business transformations belong downstream. Keeping raw data intact guarantees reproducibility and allows us to reprocess if rules change later.
8. Querying a production OLTP source for profiling or extraction can degrade the application because it adds load to a system meant for transactions. Heavy queries may slow down user operations or even cause outages.
9. If the API had a rate limit of 60 requests per minute, I would add throttling or batching logic. The pipeline would respect the limit by pausing between requests or reducing page size to stay within the allowed rate.
10. To extend this pipeline from a local raw area to PostgreSQL while preserving rerun safety, I would load the raw files into staging tables, apply deduplication and watermark logic, and then insert into final tables. The same idempotent rules would apply so reruns produce consistent results.